

Experiment No. 33

Title

Prolog Program to Find the Number of Vowels

Aim

To write a Prolog program to find the **number of vowels in a given word**.

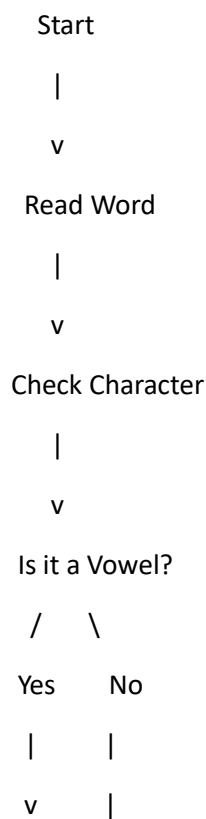
Objective

The objective of this experiment is to count the vowels present in a given list of characters using Prolog.

Algorithm

1. Start.
2. Define the vowels.
3. Read the input word as a list of characters.
4. Check each character.
5. Increment the count if it is a vowel.
6. Display the total number of vowels.
7. Stop.

Flowchart



Increase Count |

\ /
v v

More Characters?

/ \

Yes No

| |

└─► Check v

Display Count

|

v

Stop

Prolog Program

vowel(a).

vowel(e).

vowel(i).

vowel(o).

vowel(u).

count_vowels([], 0).

count_vowels([H|T], Count) :-

 vowel(H),

 count_vowels(T, C),

 Count is C + 1.

count_vowels([H|T], Count) :-

 \+ vowel(H),

 count_vowels(T, Count).

Query

?- count_vowels([h,e,l,l,o], Count).

Sample Output

Count = 2.

Result

The Prolog program successfully counted the number of vowels present in the given word.

Conclusion

Thus, the number of vowels in a given word was successfully determined using Prolog.